

# **KSA Office IT Support & Infrastructure Management**

## **Phase 07: IT Support Operations**

### **Afaq Business Solutions Co.**

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## **Network Troubleshooting Standard Operating Procedures (SOP)**

### **1. Purpose**

Provide a standardized process for diagnosing workstation network connectivity and DNS issues.

### **2. Scope**

Applies to IT Support personnel handling network connectivity incidents.

### **3. Network Incident Workflow**

- Receive and document the user's report.
- Verify network interface status.
- Check the default route.
- Test external IP connectivity.
- Test DNS name resolution.
- Document evidence and findings.
- Resolve or escalate the incident.

### **4. Connectivity Testing**

- Use `ip addr` to review interface status and assigned addresses.
- Use `ip route` to verify the default route.
- Use `ping` with a known public IP to test basic connectivity.

### **5. DNS Troubleshooting**

- Use `getent hosts` to test hostname resolution.
- Compare IP connectivity and DNS results to identify the failure layer.

## **6. Evidence Collection**

- Record relevant diagnostic results.
- Avoid unnecessary configuration changes during diagnosis.

## **7. Escalation Criteria**

- Escalate infrastructure-level problems outside the workstation.
- Escalate persistent routing, gateway, or DNS failures according to company procedures.

## **8. Security Considerations**

- Do not expose credentials or sensitive network information.
- Make only authorized network configuration changes.

## **9. Best Practices**

- Follow a layered troubleshooting approach.
- Document each diagnostic step.
- Verify service availability after troubleshooting.
- Inform the user before closing the ticket.